Des-Lux Play

Part 3: Changes and Modifications Report

Enhancing Functionality and SEO

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1. Summary

This report records every change made to the Des-Lux Play website between Part 2 and Part 3. The site stays a five-page static site. It now has JavaScript interactivity, search-engine optimisation, validated forms with AJAX submission, and is ready to deploy on GitHub Pages.

Measure	Result
HTML pages	5 original pages updated, plus a new 404.html
New JavaScript files	6 (config, site, products-data, products, forms, map), about 53 KB of source
Stylesheet	css/style.css grew from about 16 KB to about 32 KB (new section 14)
Image weight	about 23 MB of PNG reduced to 1.1 MB (WebP, resized), twelve images
New SEO files	robots.txt, sitemap.xml
Automated checks	53 browser tests passed (filters, dialogs, tabs, forms, newsletter, lightbox, counter); 0 broken links; 0 duplicate IDs; no horizontal scrolling at 390 px

Before and after (home page on desktop, then the new product catalogue):

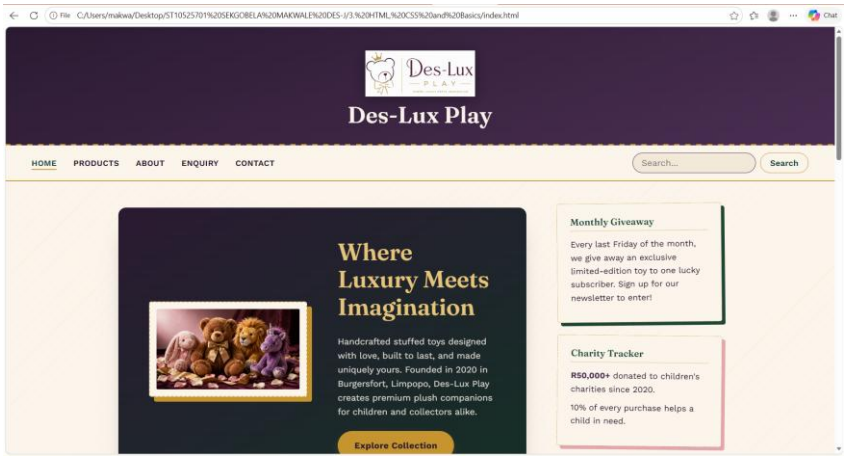


Figure 1. Part 2 home page screenshot

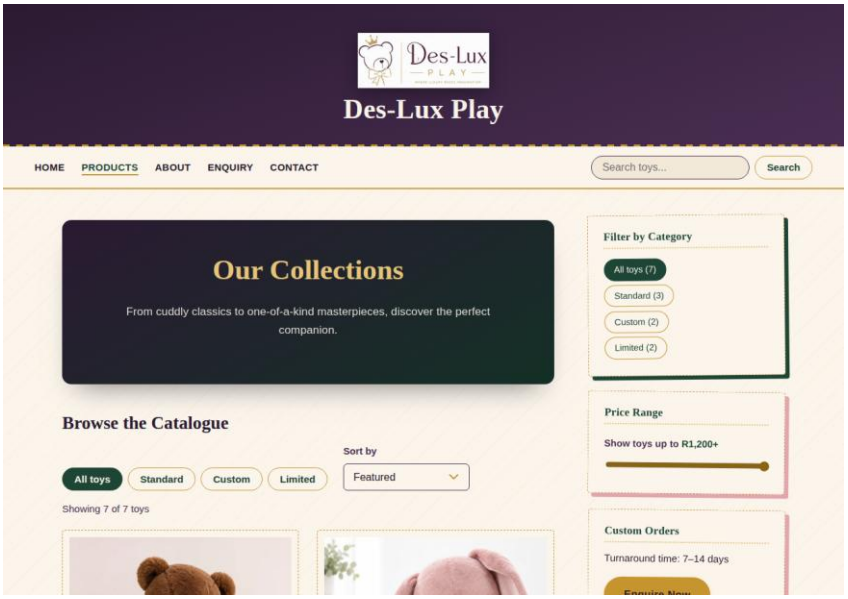


Figure 2. Part 3 products page with filters, sort and price slider

Screenshots were taken in a test browser without internet access, so Google Fonts show as the fallback serif.

2. Working through feedback from Part 2

2.1 Open items from the Part 2 README, now resolved

Issue from Part 2	Change made	Files
Search box was markup only	Now a real GET form. On products.html it filters live; on other pages it opens products.html?q=... Empty searches are blocked.	all pages, site.js, products.js
Newsletter signup was markup only	Now a form with e-mail validation, honeypot, AJAX submission and a status message.	index.html, site.js
Category filter links were href="#"	Links open products.html?category=... and filter instantly. Sidebar counts now match the data.	products.html, products.js
Forms posted to a placeholder	Forms validate and submit by AJAX. The placeholder stays until a Formspree ID is added; demo mode makes this visible.	enquiry.html, contact.html, forms.js, config.js
Map embeds were approximations	Google iframes replaced by a Leaflet map built from coordinates in the page, plus Google Maps links.	contact.html, map.js
Image filenames had typos (standdard, uicoron, fouder)	All images renamed with descriptive names and every reference updated.	images/, all pages

2.2 Further corrections found while working

Issue	Change made	Files
Brand name was the h1 on every page; h2 was skipped	One h1 per page; sections h2; sidebar headings h3. Header brand is a styled div.	all pages, style.css
Logo width/height (500x150) did not match the image	Corrected to 500x209.	all pages
Gold text had poor contrast on ivory	Added --gold-dark and used it for gold text.	style.css
Fonts loaded through CSS @import	Moved to <link> with preconnect.	all pages, style.css
README screenshot path differed from the folder tree	Paths now match; screenshots retaken.	README.md, docs/

3. Changes by file

File	Changes
index.html	Single h1 in hero; collection headings now h2; new call-to-action links to filtered products, enquiry and About; newsletter is a validated form with honeypot; charity total animates; new head with SEO, Open Graph, CSP, JSON-LD (Organization and WebSite with site search); hero image preloaded; below-the-fold images lazy-loaded; WebP images.

File	Changes
products.html	Static sections replaced by a JavaScript-built catalogue (7 toys); toolbar with category chips and sort; price slider and recently viewed in the sidebar; quick-view modal; Good to Know tabs (Sizes, Ordering and delivery, Care); noscript fallback list; ItemList JSON-LD with Product offers.
about.html	h1/h2 hierarchy fixed; in-text links to products and enquiry; new Gallery section with 10 photos and a lightbox dialog; founder image preloaded.
enquiry.html	Form rebuilt with validation attributes and data-validate rules; new Needed-by date and hints; honeypot; hidden form name; result panel (estimate, budget fit, availability, deposit, e-mail button); FAQ accordion (5 questions); FAQPage JSON-LD; ?toy= preselects the toy type.
contact.html	Form gains optional phone, minimum message length and character counter; compiled-email result panel with Open in my e-mail app button; iframes replaced by an interactive Leaflet map, a Find my nearest store button, and two location cards; Store JSON-LD for both locations; Leaflet loaded with Subresource Integrity.
404.html (new)	Friendly page for broken links on GitHub Pages; marked noindex.
css/style.css	Heading scale shifted for the new h1/h2/h3 hierarchy; @import removed; --gold-dark added; header brand and nav search restyled for the new markup; new section 14 covering wide sections, entrance animation, catalogue cards, dialogs, tabs, accordion, gallery, form errors and result panels, map and pins, footer links, responsive rules and reduced-motion handling.
README.md	Rewritten for Part 3: features, SEO documentation, security, deployment steps, before-going-live list, full changelog and references.

4. New JavaScript

File	Purpose
js/config.js	One place for the form URL, e-mail recipient, turnaround, estimate prices and deposit. Detects demo mode while the Formspree placeholder remains.
js/site.js	Shared helpers (DOM builder, e-mail check, AJAX sender); header search guard; newsletter; count-up animation; accordion; tabs; gallery lightbox.
js/products-data.js	The catalogue as a JavaScript array: 3 standard, 2 custom, 2 limited-edition toys.
js/products.js	Renders cards; live search, category filter, price filter, sort; shareable URLs; quick-view modal; recently viewed saved in localStorage.
js/forms.js	Validation rules, error display, character counters, AJAX submission, enquiry estimate and compiled-email result panels.
js/map.js	Leaflet map with OpenStreetMap tiles, stitched-pin markers, popups, Show on map buttons, nearest-store geolocation, and a fallback if the library fails to load.

4.1 Interactive elements

- Accordion on the enquiry FAQ (WAI-ARIA pattern; collapsed panels are inert; arrow keys move between questions).
- Tabs on the products page (Left, Right, Home and End keys).
- Modal quick view using a native dialog (focus trap, Esc, backdrop click).
- Lightbox gallery on About (previous, next, arrow keys, photo counter, wraps around).

- Interactive map on Contact (Leaflet and OpenStreetMap).
- Animations: banner photograph settles into its frame on load; accordion height transition; charity total counts up when scrolled into view. All disabled under prefers-reduced-motion.
- Advanced DOM manipulation: element builder, DocumentFragment, replaceChildren, one delegated click listener, IntersectionObserver; visitor text inserted with textContent only.

5. Forms and validation

	enquiry.html	contact.html
Fields	Name, e-mail, phone, toy type, size, colours, budget, description, needed-by date, reference image	Name, e-mail, phone (optional), message type, message
After a valid submit	Estimate (indicative price, budget fit, availability, 50% deposit) and an E-mail this enquiry button	Compiled e-mail (to, subject, body) and an Open in my e-mail app button
Recipient	info@desluxplay.co.za (set in js/config.js)	info@desluxplay.co.za (set in js/config.js)

- Names: letters, spaces, hyphens and apostrophes only, 2 to 60 characters.
- E-mail: format check. Phone: South African format (082 123 4567 or +27 82 123 4567).
- Needed-by date: not sooner than 7 business days ahead.
- Reference image: must be an image, 2 MB or smaller.
- Messages: length limits with a live character counter.
- Errors appear under each field, are linked with aria-describedby, mark the field aria-invalid, and focus moves to the first error.
- AJAX: fetch POST with a Sending state and error handling; without JavaScript the form still posts normally.

6. Search engine optimisation

6.1 Titles and descriptions

Page	Title	Meta description (summary)
Home	Des-Lux Play | Luxury Handmade Stuffed Toys in Limpopo	Luxury handmade toys, standard, custom and limited edition, 10% to charity
Products	Luxury Plush Toys & Custom Teddies | Des-Lux Play	Handmade bears, custom plush, limited editions; prices from R300
About	Our Story & Charity Work | Des-Lux Play	Founder story and the 10% charity pledge
Enquiry	Custom Stuffed Toy Enquiry & Quote | Des-Lux Play	Instant estimate; made in 7-14 business days
Contact	Contact & Store Locations | Des-Lux Play	Both stores, hours, phone numbers and interactive map

6.2 Other SEO work

- Keyword research table in the README (candidate keywords per page; search volume not measured).

- Heading structure: one h1 per page.
- Descriptive image file names and alt text.
- Internal linking in page copy, calls to action and a footer menu on every page.
- Mobile-friendly layout confirmed with no sideways scrolling at 390 px.
- Open Graph and Twitter card tags, canonical URLs, robots meta.
- JSON-LD structured data: Organization, WebSite, ItemList of Products, FAQPage and two Stores.
- robots.txt and sitemap.xml added.
- Off-page plan (backlinks, social media, local SEO and Google Business Profile) documented in the README as actions for the owner.

7. Performance and security

7.1 Image optimisation

Original file	New file	Before	After
hero-banner.png	luxury-handmade-stuffed-toys-hero.webp	2.3 MB	114 KB
charity-kids.png	des-lux-play-charity-donation-children.webp	2.2 MB	93 KB
product-standdard.png	standard-collection-chestnut-teddy-bear.webp	2.0 MB	77 KB
product-custom-uicoron.png	custom-purple-unicorn-plush.webp	2.2 MB	79 KB
product-custom.png	custom-emerald-dragon-plush.webp	2.1 MB	93 KB
product-limited.png	limited-edition-golden-lion-plush.webp	2.3 MB	114 KB
about-fouder.png	des-j-sekgobela-founder-stitching.webp	2.3 MB	128 KB
logo.png	des-lux-play-logo.webp	274 KB	9 KB
favicon.png	favicon.png (resized)	330 KB	42 KB
Velvet Bunny.png	velvet-bunny-pastel-pink-plush.webp	2.2 MB	about 80 KB
Cuddly Elephant.png	cuddle-elephant-grey-plush.webp	2.4 MB	about 95 KB
Silver Fox.png	silver-fox-limited-edition-plush.webp	2.3 MB	85 KB

- Other speed work: fonts load through preconnect with display swap; scripts are deferred; lazy loading below the fold; hero preload; width and height on every image.
- Security: Content Security Policy and referrer policy on every page; Subresource Integrity on the Leaflet files; honeypot field on every form; length limits; no visitor text inserted as HTML; rel=noopener on external links.
- A meta tag cannot set every security header, and GitHub Pages does not allow custom headers.

8. Repository and deployment

- Site files moved to the repository root so GitHub Pages can serve index.html.
- New folder layout: css/, js/, images/, docs/screenshots/.
- Deployment steps are in the README (Settings, Pages, Deploy from branch, main, root).

- After deploying, replace the placeholder address in the canonical and og tags, JSON-LD, robots.txt, sitemap.xml and the README.

9. Still to do before submission

Item	Why
Add Part 2 lecturer feedback to sections 2 and the README changelog	Not available when this was written
Create a Formspree form and replace YOUR_FORM_ID (js/config.js and three action attributes)	Until then forms run in demo mode and send nothing
Confirm estimate prices in js/config.js	Only the R500 starting price came from the brief; larger sizes are placeholders
Replace placeholder phone numbers, e-mails, addresses and map coordinates	Details are stand-ins; coordinates are approximate town centres
Check the live map after deploying	Leaflet loads from the internet and could not be tested live; the map logic was tested with a stand-in library
Add access dates to the README references	Required by the referencing style
Push to GitHub, enable Pages, submit the repository link	Submission requirement